

TECHNICAL PRESENTATION & STUDY GUIDE

Machine Learning for Industrial Corrosion Rate Prediction

A Comprehensive Guide explaining Preprocessing, Models, Evaluation, and Visualizations

Project Domain: Asset Integrity Management & Predictive Maintenance

Machine Learning Framework: Scikit-Learn & XGBoost Regressor

Target Variable: corrosion_rate_mpy (Mils Per Year)

Dataset Scale: 100,000 Industrial Assets

Prepared for: Technical Presentation to Project Advisor / Professor

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1. Introduction & Problem Context

Industrial corrosion is a multi-billion dollar engineering challenge, causing asset degradation, unexpected failures, environmental hazards, and safety risks. Traditionally, asset integrity engineers rely on empirical formulas, coupon exposures, or periodical inspections to estimate corrosion rates. However, these methods are either retrospective or struggle to model the complex, non-linear interactions of environmental severity, material resistance, and operational pressures.

Objective: This project develops a machine learning pipeline to predict the corrosion rate (expressed in *mils per year* (*mpy*)) of industrial assets. By modeling these rates as a function of environmental severity, chemical composition, material choice, coating type, and maintenance schedules, operators can transition from reactive inspections to a proactive, predictive maintenance regime.

Dataset Overview: The dataset consists of **100,000 distinct asset records** with 33 variables. It represents a synthetic but highly realistic engineering dataset, including categorical parameters (service environment, material alloy, protective coating type, maintenance type, mitigation barriers) and numeric parameters (pH, chloride levels, temperature, operating pressure, pipe diameter, wall thickness, exposed surface area, exposure time, inhibitor concentrations, and safety risk scores). The target variable is **corrosion_rate_mpy**, representing the annual wall loss rate.

Key Presentation Pitch: "Instead of treating corrosion as a linear time-dependent function, our model captures multi-dimensional interaction effects. For instance, how pH and chloride levels interact with specific material alloys (like Carbon Steel vs. Ti Grade 2) under high temperatures. This enables precise asset-by-asset risk indexing rather than using generic lookup tables."

2. Data Preprocessing & Feature Engineering

Data cleaning and formatting are vital to ensure model stability and accuracy. The following preprocessing steps were applied to the raw dataset:

A. Handling Missing Values:

A critical engineering assessment was performed on missing data. We observed that **coating** had 14.2% missing values, and **maintenance** had 54.7% missing values. Rather than using mechanical imputation (like filling with the mode or median), we recognized that a missing value in these columns carries physical meaning: **it represents the absence of a coating or the absence of an active maintenance control**. Therefore, we imputed all NaNs in these columns with the string 'None'. This explicitly forces the machine learning models to learn the baseline degradation rate of untreated or unmaintained assets.

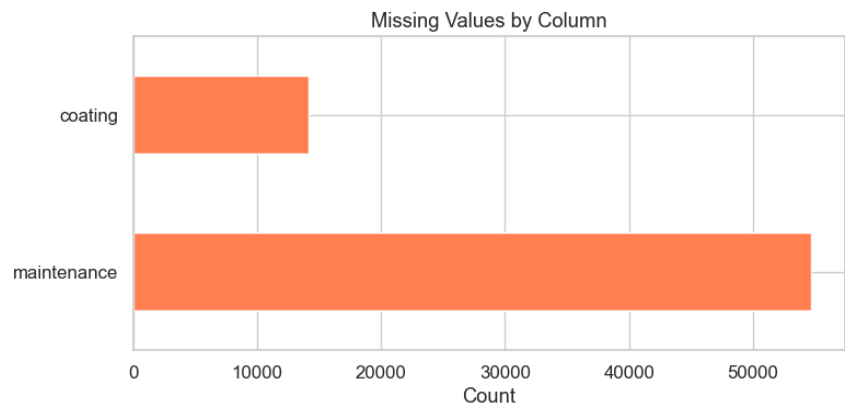


Figure 1: Missing Value Analysis showing null values in 'maintenance' (54.7%) and 'coating' (14.2%). All other features are 100% complete.

B. Categorical Encoding:

Machine learning models require numerical inputs. The categorical features (environment, material, coating, maintenance) are nominal and have low cardinality (between 4 and 8 unique categories). We applied **One-Hot Encoding** using Pandas `get_dummies()`. This transformed the 4 categorical columns into 23 binary indicator variables (e.g. material_Carbon Steel, environment_CO2 Pipeline Wet), increasing the total feature count to 42. Since cardinality was low, this did not cause dimension-related overfitting (often called the curse of dimensionality).

C. Feature Scaling:

Features like pressure, temperature, and pipe diameter operate on vastly different numerical scales. For **Linear Regression**, we applied a **StandardScaler**, which centers the features to a mean of 0 and scales them to a standard deviation of 1. Scaling prevents variables with large absolute ranges (e.g., total cost, diameter in mm) from dominating the cost function and coefficient estimations. *Note: Tree-based models (Random Forest and XGBoost) do not require feature scaling because they split nodes based on monotonic ordering, making them invariant to scale.*

D. Data Partitioning (Train/Val/Test Split):

To ensure unbiased evaluation, the dataset was partitioned into **Training (70%)**, **Validation (15%)**, and **Testing (15%)** sets. We utilized the pre-allocated split column from the original database to ensure exact reproducibility across all models. The training set fits the model parameters, the validation set is used for tuning hyperparameters and early stopping (specifically in XGBoost), and the test set acts as an completely unseen holdout set for final evaluation.

3. Exploratory Data Analysis & Core Findings

Exploratory Data Analysis (EDA) revealed key engineering trends, correlations, and distributions that guided our model development.

A. Target Distribution Analysis:

The target variable `corrosion_rate_mpy` is heavily skewed. It exhibits a **skewness of 4.335** and a **kurtosis of 31.539**. This positive skew indicates that the vast majority of assets experience low, manageable corrosion rates (median below 1.0 mpy), but a small subset of assets experience extreme, catastrophic corrosion rates exceeding 20 mpy. In statistical modeling, this indicates a heavy-tailed distribution, meaning standard linear models will suffer because they minimize mean squared errors, pulling the fit towards extreme outliers at the expense of general prediction accuracy. Tree-based ensembles are much better suited to capture these localized, high-corrosion regimes.

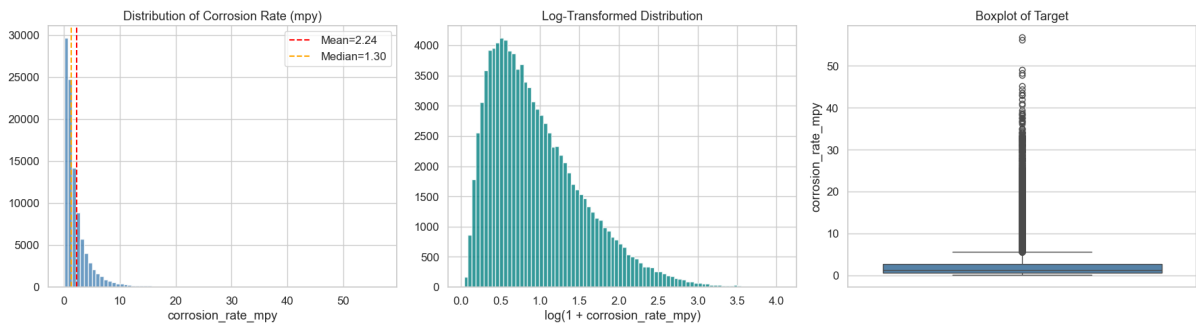


Figure 2: Distribution of target variable (`corrosion_rate_mpy`) showing a highly skewed, heavy-tailed distribution.

B. Environmental Severity Ranking:

The service environment is the single most critical factor in determining degradation rates. The data reveals a clear ranking of environmental severity:

- **CO2 Pipeline Wet:** Mean corrosion rate of **5.48 mpy** (Highest risk — acidic carbonic acid formation).
- **Process Brine:** Mean corrosion rate of **3.56 mpy** (High salinity drives electrochemical reactions).
- **Marine Splash Zone:** Mean corrosion rate of **2.56 mpy** (High oxygen + chloride exposure).
- **Industrial Urban:** Mean corrosion rate of **1.95 mpy** (Atmospheric sulfur and nitrogen oxides).
- **Marine Immersed:** Mean corrosion rate of **1.70 mpy** (Fully submerged salt-water corrosion).
- **Soil Clay / Sandy / Rural:** Low risk environments (ranging from **0.78 to 1.07 mpy**).

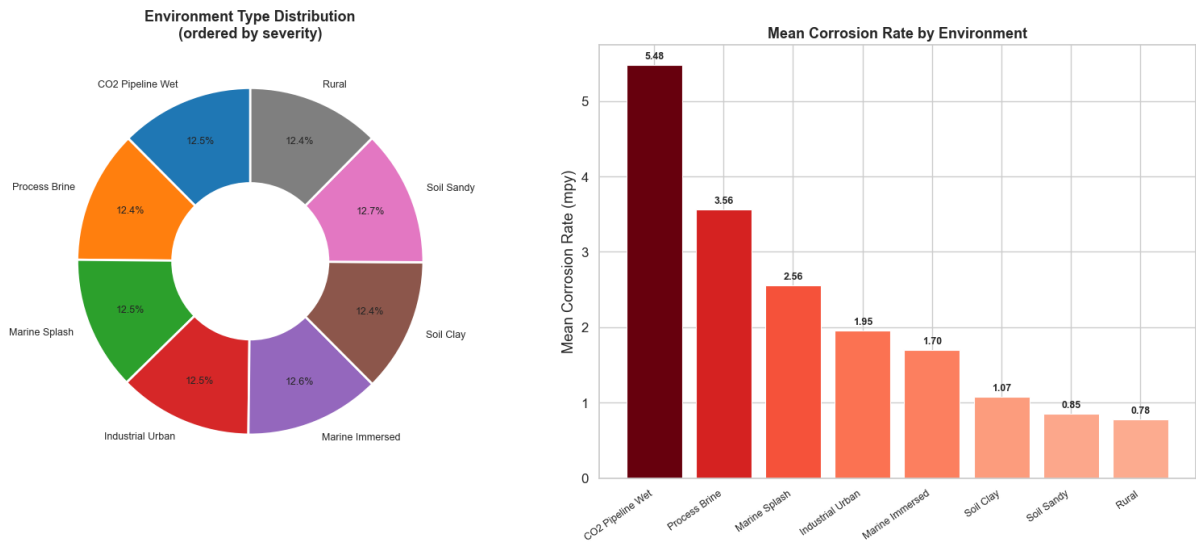


Figure 3: Service Environment Distribution and associated Mean Corrosion Rates (MPY). Wet CO2 is the most severe class.

C. Protective Coating Impact:

Applying protective coatings acts as the primary barrier against the external environment. Uncoated assets (coating = 'None') exhibit an alarmingly high mean corrosion rate of **5.28 mpy**. The analysis confirms the relative effectiveness of different coating technologies:

- **3-Layer Polyethylene (3LPE):** Most effective, restricting average corrosion to **1.08 mpy** (a 79.5% reduction vs. uncoated).
- **Fusion Bonded Epoxy (FBE):** Highly effective, holding corrosion to **1.32 mpy**.
- **Zinc-Aluminum Metallizing:** Restricts corrosion to **1.59 mpy**.
- **Coal Tar / Wrap Tape / Paint Epoxy:** Moderately effective, with mean rates ranging from **1.83 to 2.41 mpy**.

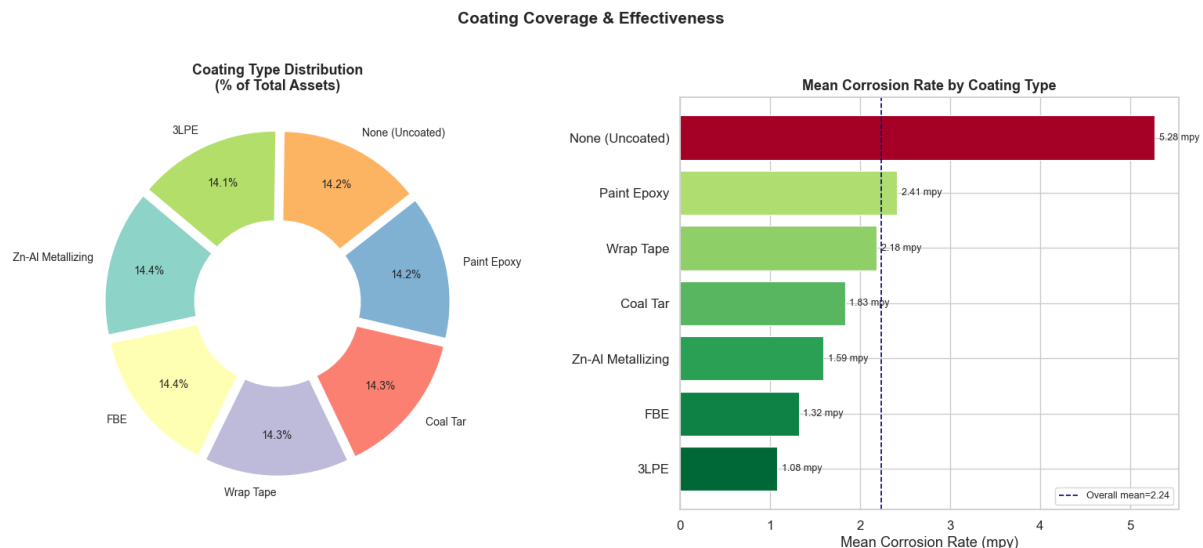


Figure 4: Coating Type Distribution and average mitigation performance relative to the uncoated baseline.

D. Material Selection and Performance:

Alloy chemistry determines the baseline resistance of the metal structure. Carbon Steel is the most vulnerable and widely used material, exhibiting a mean corrosion rate of **3.96 mpy**. Upgrading materials shows exponential improvements in durability:

- **Titanium (Ti) Grade 2:** Outstanding performance, limiting average corrosion to **0.85 mpy** (78.5% reduction vs. Carbon Steel).
- **Duplex 2205:** High-performance alloy, limiting corrosion to **1.41 mpy**.
- **Stainless Steel 316 / 304:** Restricts corrosion to **1.70 mpy** and **2.06 mpy** respectively.
- **Copper-Nickel (Cu-Ni) 90-10:** Averages **2.05 mpy**.
- **Low Alloy Steel:** Averages **3.50 mpy**, offering minor improvement over Carbon Steel.

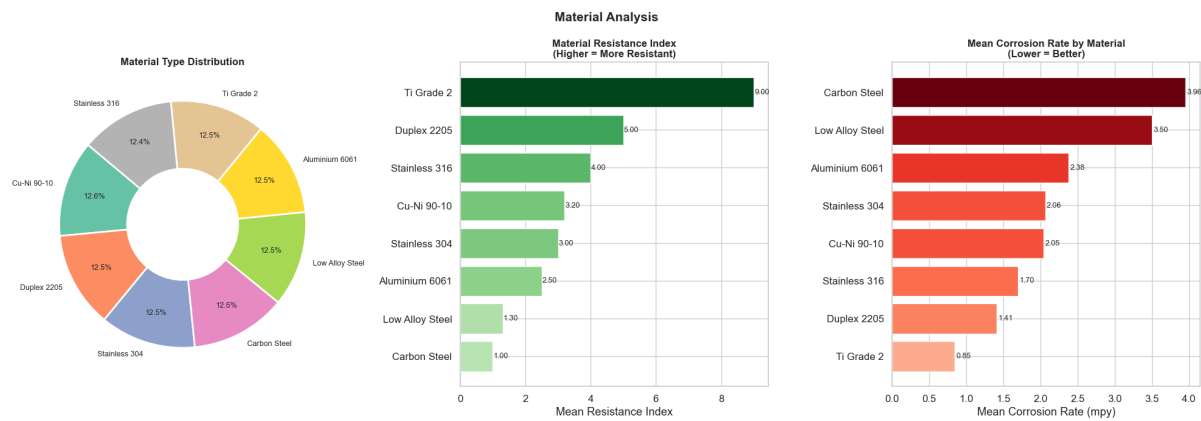


Figure 5: Material Distribution, Material Resistance Index (MRI), and average corrosion rates per alloy type.

4. Machine Learning Models & Methodology

We evaluated three distinct machine learning algorithms to model the corrosion rate. Each represents a different approach to handling tabular dataset characteristics:

1. Linear Regression (Baseline):

Linear Regression fits a linear equation to the scaled training data by minimizing the sum of squared residuals. While highly interpretable, it assumes that features have additive, linear relationships with the target and that there are no interaction terms unless explicitly hand-coded. In our pipeline, it achieved a modest **R² of 62.3% on the test set**. The high average error (MAE = 1.02 mpy) and massive percentage error (MAPE = 117.6%) indicate that a simple linear assumption is inadequate for modeling complex corrosion chemical kinetics.

2. Random Forest Regressor:

Random Forest is a bagging ensemble method that trains multiple independent decision trees in parallel. It uses bootstrap aggregating (bagging) and random feature selection to reduce variance and avoid overfitting. By splitting nodes on localized conditions, it naturally captures non-linear relationships and complex interactions (e.g. temperature's effect is dependent on pH). Random Forest dramatically outperformed the linear baseline, reaching an **R² of 87.8% on the test set** and reducing the MAPE down to **24.2%**.

3. XGBoost Regressor (Final Chosen Model):

XGBoost (Extreme Gradient Boosting) is an optimized distributed gradient boosting library. Unlike Random Forest, it trains decision trees sequentially. Each new tree is fit to predict the residuals (errors) of the preceding trees, using a gradient descent algorithm. It incorporates L1 and L2 regularization to penalize complex trees and prevent overfitting. In our pipeline, we configured XGBoost with n_estimators=500, max_depth=8, and a conservative learning_rate=0.05, utilizing the validation set for early stopping to halt training if the validation loss stopped improving. XGBoost achieved the highest performance, with a **test R² of 88.7%**, **MAE of 0.4746 mpy**, and **MAPE of 22.8%**.

Why XGBoost was Selected: XGBoost offers three major advantages for this application: (1) Superior accuracy: it achieved the lowest MAE and highest R². (2) Robustness to non-linear chemical kinetics: it requires no manual feature interactions. (3) Numerical efficiency: gradient boosting with histograms is exceptionally fast, and it has built-in mechanisms to handle outliers and raw features without scaling, simplifying deployment.

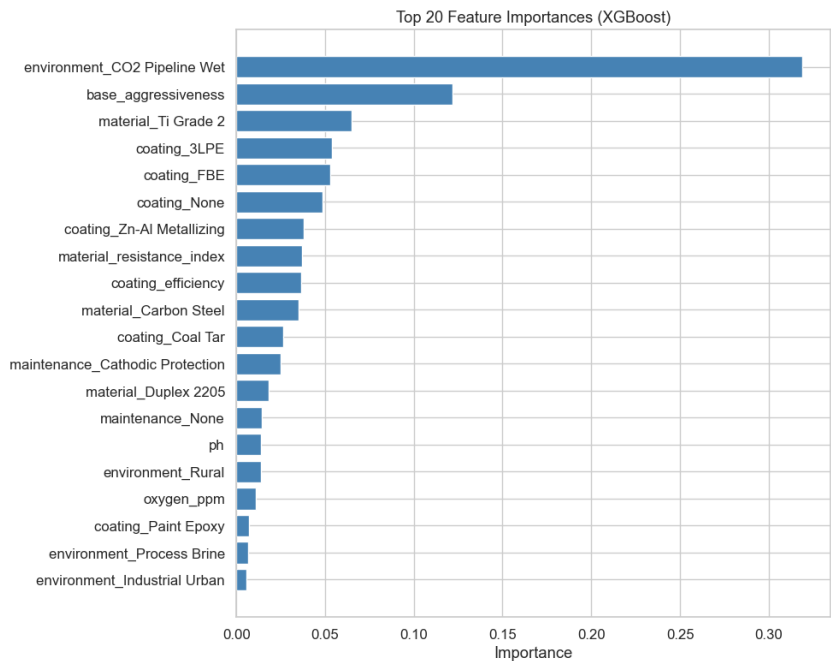


Figure 6: Model Feature Importance ranking from the final XGBoost regressor, showing environmental indicators dominating.

5. Performance Evaluation & Metrics

To evaluate model accuracy, we utilize four standard statistical metrics. The mathematical formulas and their specific physical interpretations in our engineering context are outlined below:

- **Mean Absolute Error (MAE):** Calculates the average of the absolute differences between predictions and actual values. Its physical unit is *mpy*. An MAE of 0.4746 means that, on average, the model's predicted corrosion rate deviates from the actual rate by just under 0.5 mils per year. This represents an exceptionally tight window of error.
- **Root Mean Squared Error (RMSE):** Computes the square root of the mean squared differences. Because it squares the errors before averaging, it heavily penalizes large, localized errors. This is critical in safety-sensitive applications: underestimating a high corrosion rate is far more dangerous than overestimating it. The XGBoost model's RMSE of 0.9838 mpy indicates that extreme errors are successfully constrained.
- **Coefficient of Determination (R²):** Measures the proportion of variance in the target variable that is explained by the features. An R² of 88.7% indicates that the model captures nearly 90% of the underlying physical and environmental drivers of corrosion, leaving only 11.3% to random noise or unmeasured variables.
- **Mean Absolute Percentage Error (MAPE):** Shows the average percentage deviation of the predictions. While helpful, it can become artificially inflated when actual values are very close to zero (as dividing by a tiny actual value yields a massive percentage). This explains the baseline Linear Regression's MAPE of 117.6%, which was successfully reduced to 22.8% by XGBoost.

Model Performance Summary Table:

Model Evaluation Set	MAE (mpy)	RMSE (mpy)	R ² (%)	MAPE (%)
Linear Regression (Val)	1.0230	1.7415	63.09%	114.47%
Linear Regression (Test)	1.0201	1.7996	62.30%	117.58%
Random Forest (Val)	0.4844	0.9497	89.03%	24.30%
Random Forest (Test)	0.4985	1.0247	87.78%	24.17%
XGBoost (Val) — Champion	0.4617	0.9110	89.90%	22.62%
XGBoost (Test) — Champion	0.4746	0.9838	88.73%	22.83%

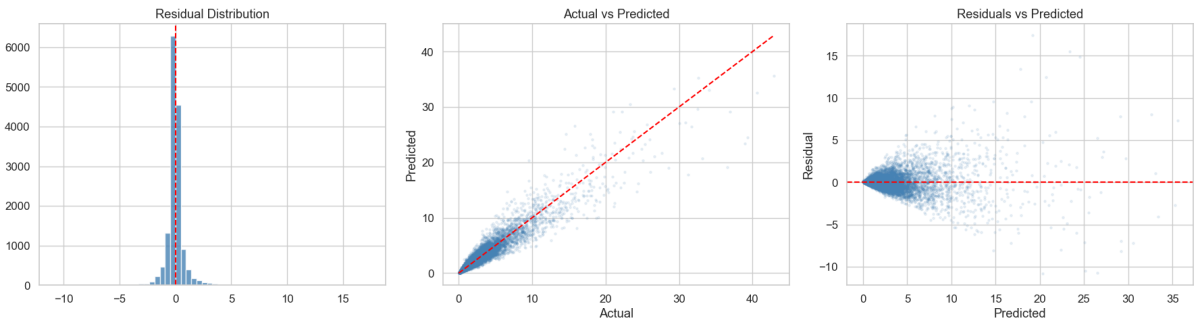


Figure 7: Residual Analysis for XGBoost, demonstrating a normally distributed residual distribution, high linearity in the Actual vs. Predicted plot, and stable variance (homoscedasticity).

6. Graph & Visualization Interpretation Guide

This section acts as a cheat sheet for explaining each figure in the notebook during your presentation. It provides the statistical purpose, physical meaning, and the exact conclusion you should present.

A. Correlation Heatmap (Figure 8)

What is plotted: A Pearson correlation matrix of all numeric features against the target variable, corrosion_rate_mpy.

Physical Meaning: It measures the linear strength and direction of relationships. Red cells denote positive correlation (as feature increases, corrosion increases), and blue cells denote negative correlation.

Presentation Pitch: "The correlation matrix shows that temperature, ph, base aggressiveness, and chloride concentrations have the strongest relationships with corrosion. However, because correlation only measures linear relationships, it underestimates the impact of categorical combinations. This is why our non-linear models achieve far higher accuracy than linear regression, as they capture interactions that the correlation matrix misses."

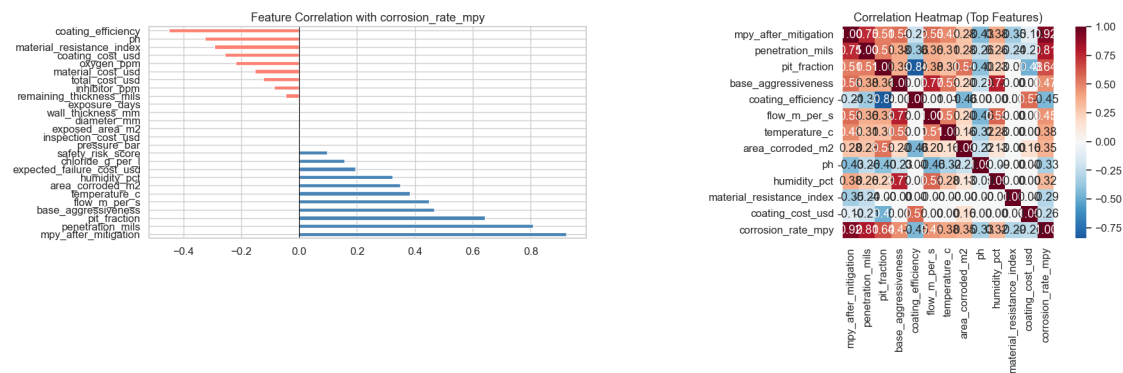


Figure 8: Correlation Matrix of numerical operational parameters against target variable.

B. Material Performance Heatmaps (Figures 9 & 10)

What is plotted: Cross-tabulated grids showing the average corrosion rate for each Material alloy across different Environments (Figure 9) and Coating types (Figure 10).

Physical Meaning: This visualizes critical engineering interaction effects. For instance, Carbon Steel in Rural environments has low corrosion, but in Wet CO2 pipelines it experiences catastrophic rates. It helps identify exactly which material-environment-coating combinations are safe.

Presentation Pitch: "These heatmaps demonstrate why material selection must be tailored to the environment. For example, in a Wet CO2 Pipeline, Carbon steel exhibits an average rate of over 10 mpy, which would cause wall breach in a few years. Upgrading to Titanium Grade 2 or Duplex 2205 reduces this rate to under 1.5 mpy, offering a permanent metallurgical solution. Similarly, combining Carbon Steel with a 3LPE coating successfully mitigates the aggressive environment without requiring expensive alloy upgrades, demonstrating the economic trade-offs we can model."



Figure 9: Material x Environment Heatmap, plotting mean corrosion rates (mpy). Note the extreme vulnerability of Carbon Steel in Wet CO₂.



Figure 10: Material x Coating Heatmap, plotting mean corrosion rates (mpy). Demonstrates the high synergistic performance of 3LPE and FBE coatings across all alloys.

C. Cost Analysis & Lifecycle Economics (Figure 11)

What is plotted: A breakdown of the average total cost (\$1,868.84 per asset) into Material cost (48.6%), Coating cost (8.9%), Inspection cost (6.0%), and Expected Failure cost (36.5%). It also features a scatter plot of Corrosion Rate vs. Total Cost and a box plot of Total Cost by Environment.

Physical Meaning: This links machine learning predictions to financial decisions. High corrosion rates lead to a high probability of failure, dramatically increasing the 'Expected Failure Cost' (which is the financial consequence of failure multiplied by the probability of failure). Conversely, using high-grade alloys or expensive coatings reduces failure probability but increases initial capital expenditure (CAPEX).

Presentation Pitch: "A key finding is that assets in the bottom 10% of corrosion rates actually have a HIGHER mean total cost (\$2,298) compared to assets in the top 10% of corrosion rates (\$1,597). This sounds counterintuitive but represents a classic engineering trade-off: to achieve low corrosion, operators must invest heavily in expensive alloys (Ti Grade 2, Duplex 2205) and advanced coatings, raising initial CAPEX. On the other hand, accepting moderate corrosion rates by using cheaper Carbon Steel and focused maintenance can optimize the total lifecycle cost, provided the risk of failure is kept within safety margins. Our model acts as an optimization engine to balance CAPEX vs. OPEX."

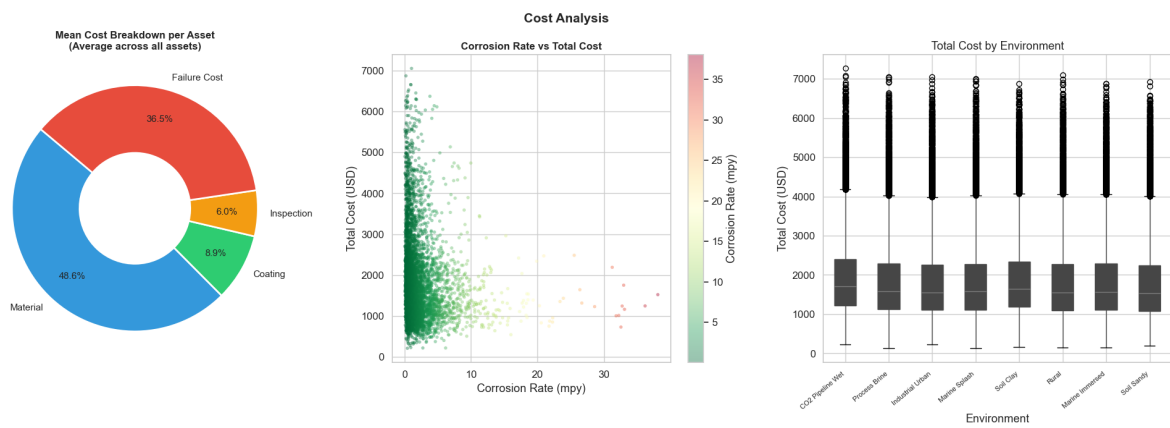


Figure 11: Lifecycle Cost Analysis showing breakdown (donut), corrosion vs. total cost (scatter), and cost distributions per environment (box).

D. Safety Risk Score & Asset Exposure Analysis (Figure 12)

What is plotted: Distribution of Safety Risk Scores, scatter plot of Corrosion Rate vs. Safety Risk, and Exposure Duration analysis.

Physical Meaning: Safety Risk represents a composite index of operating pressure, temperature, location hazard (e.g. proximity to population), and wall thickness. High corrosion rates in high-risk zones present unacceptable threats. The exposure analysis shows how corrosion progresses over time.

Presentation Pitch: "Our model demonstrates that exposure duration has a near-zero correlation (-0.003) with the instantaneous corrosion rate. This is an important scientific distinction: while total metal loss (penetration depth) increases with time, the rate of corrosion (mpy) is governed by instantaneous thermodynamic and chemical conditions (pH, temperature, flow velocity, coating degradation state). This proves that static time-based inspections are inefficient; we must inspect based on operational severity predicted by our ML model."

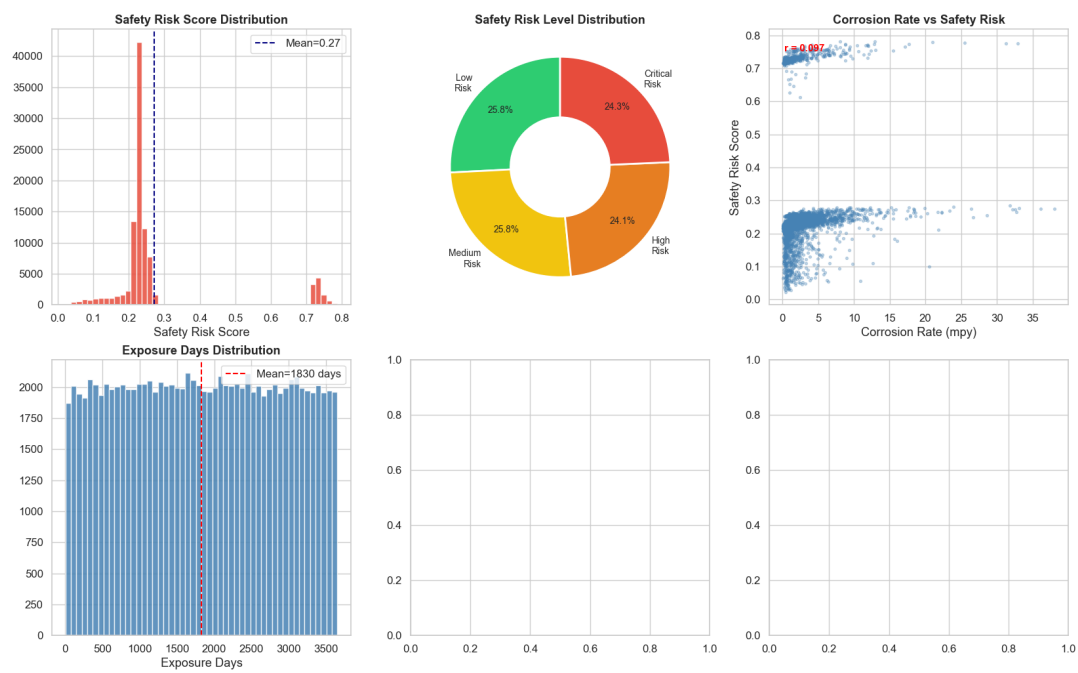


Figure 12: Safety Risk Score distributions and Exposure time correlation analysis.

7. Interactive Dashboard Overview

To make these ML insights accessible to operators and non-data-scientists, we deployed a web-based, interactive dashboard. The dashboard is built using standard, high-performance web technologies:

- **Frontend Architecture:** Structured in HTML5, styled with modern Glassmorphic CSS (using backdrop filters, custom color variables, and responsive grid layouts), and driven by vanilla JavaScript.
- **Data Engine:** Configured with a dedicated data.js file generated by our Python pipeline. It contains serialized model weights, correlations, and test predictions, ensuring instant local loading without requiring a heavy backend database during demonstrations.
- **Visualizations:** Uses Chart.js to render interactive charts for actual vs. predicted rates, feature importances, coating performance, and maintenance comparisons. Toggles allow the user to switch dynamically between the Validation and Test sets.
- **Actionable Panel:** Integrates a dedicated engineering advice card that translates model parameters into direct operating guidelines.

Presentation Demo Tip: Open the dashboard at <http://localhost:8000/dashboard/index.html>. Show the professor the interactive actual vs. predicted plot, hover over points to display individual asset predictions, and use it as a visual aid to explain the difference between 3LPE coatings and uncoated baselines.

8. Actionable Recommendations & Presentation Prep Q&A;

To help you stand out, here is a summary of actionable engineering recommendations derived from the model, followed by the top questions your professor is likely to ask and how you should answer them.

A. Engineering Action Plan:

- **Upgrade High-Severity Zones:** Wet CO2 pipelines and Process Brine lines are high-risk. Carbon steel must be avoided. Upgrade to **Titanium Grade 2** or **Duplex 2205**.
- **Optimize Coating Schedules:** Uncoated assets corrode at 5.28 mpy. Standardizing on **3LPE** or **FBE** coatings is the most cost-effective way to suppress corrosion to <1.32 mpy.
- **Enhance Maintenance Protocols:** Prioritize **Cathodic Protection** over wash intervals, as it reduces corrosion rates by more than 40% compared to basic wash schedules.
- **Targeted Chemical Dosing:** Adjust corrosion inhibitor concentrations in response to sudden changes in temperature or pH, using the ML model to predict the optimal dosing level.

B. Anticipated Q&A; Prep:

Q1: Why did you choose XGBoost over Random Forest or a Deep Learning model?

A1: XGBoost uses gradient boosting, training trees sequentially to correct the errors of previous trees, whereas Random Forest trains trees independently. This sequential focus yields higher accuracy (R^2 of 88.7% vs. 87.8%). Deep learning is overkill for tabular data of this scale, harder to tune, lacks direct feature importance interpretability, and requires significantly more computational resources for training and deployment.

Q2: Why did you one-hot encode the categorical variables instead of label encoding them?

A2: Label encoding assigns an arbitrary integer (1, 2, 3...) to categories. For nominal variables (like coating type or environment), this mistakenly implies an ordinal ranking (e.g., that 'Paint Epoxy' is mathematically 'greater than' 'FBE'), which confuses models like Linear Regression. One-hot encoding creates binary flags, treating each category as an independent feature. Because cardinalities were low (<8), one-hot encoding did not inflate dimensions.

Q3: Your Linear Regression model has a MAPE of 117%, but an MAE of 1.02. Why is the MAPE so high?

A3: This is a classic statistical artifact. MAPE divides the absolute error by the actual target value. In our dataset, many assets in mild environments have actual corrosion rates very close to zero (e.g. 0.05 mpy). If the linear model predicts 0.15 mpy (an absolute error of only 0.10 mpy), the percentage error for that single prediction is 200%. When averaged across the dataset, these near-zero divisions inflate the MAPE. MAE is a more robust indicator of absolute physical error.

Q4: Why did you split the dataset using a 'split' column rather than random train_test_split?

A4: Using a pre-allocated split column ensures exact reproducibility. If we run the script multiple times or compare different models across notebooks and files, we must evaluate on the exact same test instances. Random splits can introduce slight variations or, if not carefully seeded, lead to data leakage if features are engineered using data-dependent stats before splitting.